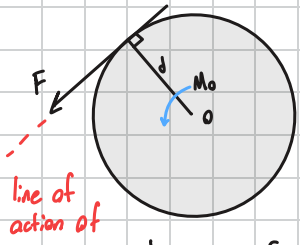
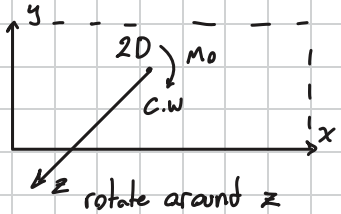
Week 7

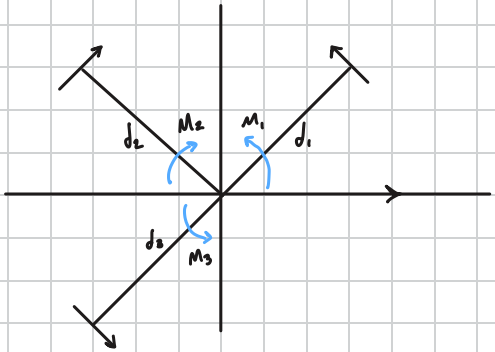
Compute moment of force in 2D

$$M_O = F \cdot d$$





direction of moment will be C.W or C.C.W

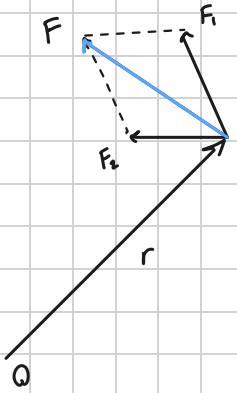


Scalor notation

$M_1 = F_1 d_1$

$M_2 = -F_2 d_2$

$M_3 = F_3 d_3$



$\vec{F} = \vec{F}_1 + \vec{F}_2$

$\vec{M}_O = \vec{r} \times \vec{F}$

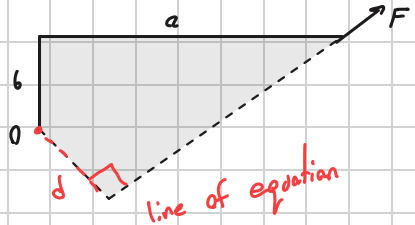
$\vec{M}_O = \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2$

$= \vec{r} \times (\vec{F}_1 + \vec{F}_2)$

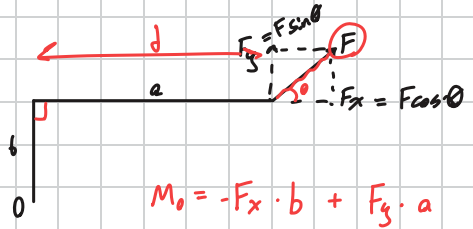
$= \vec{r} \times \vec{F}$

$M_o = F \cdot d$



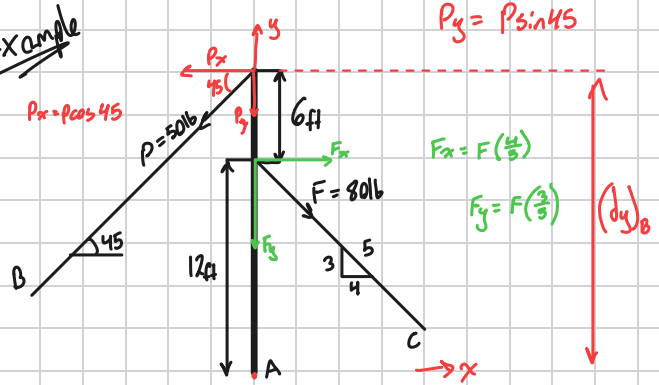


instead do this →



$M_o = -F_x \cdot b + F_y \cdot a$

Example



$P_y = P\sin 45$

$F_x = F(\frac{4}{5})$

$F_y = F(\frac{3}{5})$

$(d_y)_B$

$(M_B)_A = (P\cos 45)(12+6) + (P\sin 45)(0)$

$= 636$ lb·ft ↺ c.c.w

$(M_C)_A = -F_x(12) + F_y(0)$

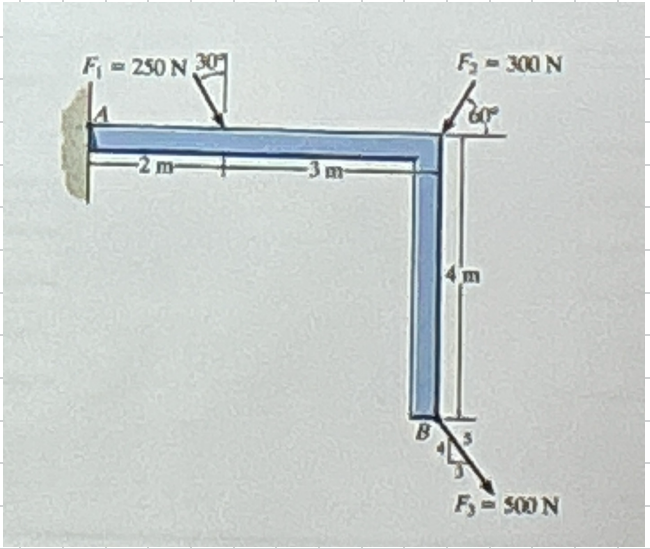
$= -F(\frac{4}{5})(12)$

$= -786$ lb·ft ↺ c.c.w

$= 786$ lb·ft ↻ c.w

pole will rotate C.W

$(M_B)_A = ?$ $(M_C)_A = ?$


$F_1 = 250$ N
30°
$F_2 = 300$ N
60°
A
2 m
3 m
4 m
B
4
5
3
$F_3 = 500$ N